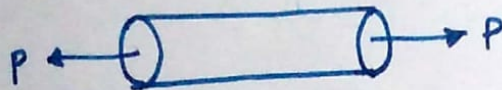


UNIT-1

Basic Equations of Linear Elasticity

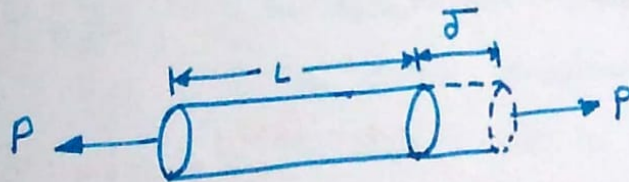
The most fundamental concepts in mechanics of materials are stress and strain.

Let us now consider a bar under uniaxial tension by a force 'P'.



The force 'P' in tension will induce in the material a stress denoted by ' σ ' (Sigma) which is nothing but the reaction to the applied pressure (-ve). Hence, we can say that the force 'P' acting over the cross-sectional area 'A' ($\pi d^2/4$) in this case induces a equal stress in the material of the bar which is ' σ '. Hence, $\sigma = P/A$.

Now, due to the load 'P' the bar changes length, as shown.



The elongated portion is the change in length (δ).

Now, we can then introduce strain which is nothing but change in length divided by the original length.

Hence, strain which is denoted by ' ϵ ' $\rightarrow$ Epsilon can be written as, $\epsilon = \frac{\delta}{L}$.

Hence, by hook's law, we can say, $\sigma \propto \epsilon$ or $\sigma = E \epsilon$

where E is the Modulus of Elasticity or Young's Modulus.

Hence, we can substitute σ & ϵ in $\sigma = E \epsilon$

$$\sigma = P/A \text{ and } \epsilon = \frac{\delta}{L} \therefore \frac{P}{A} = E \frac{\delta}{L}$$

hence, to find the deflection of a bar under Uniaxial loading we can rewrite the above equation as,

$$\frac{PL}{AE} = \delta.$$

True Stress-strain & Engineering stress strain Curves

In Engineering we always discuss the engineering stress and not the true stress.

We always divide by the Original area.

But when we consider a ductile material, which elongates and the cross sectional area decreases as a result of the elongation the stress strain curve changes.

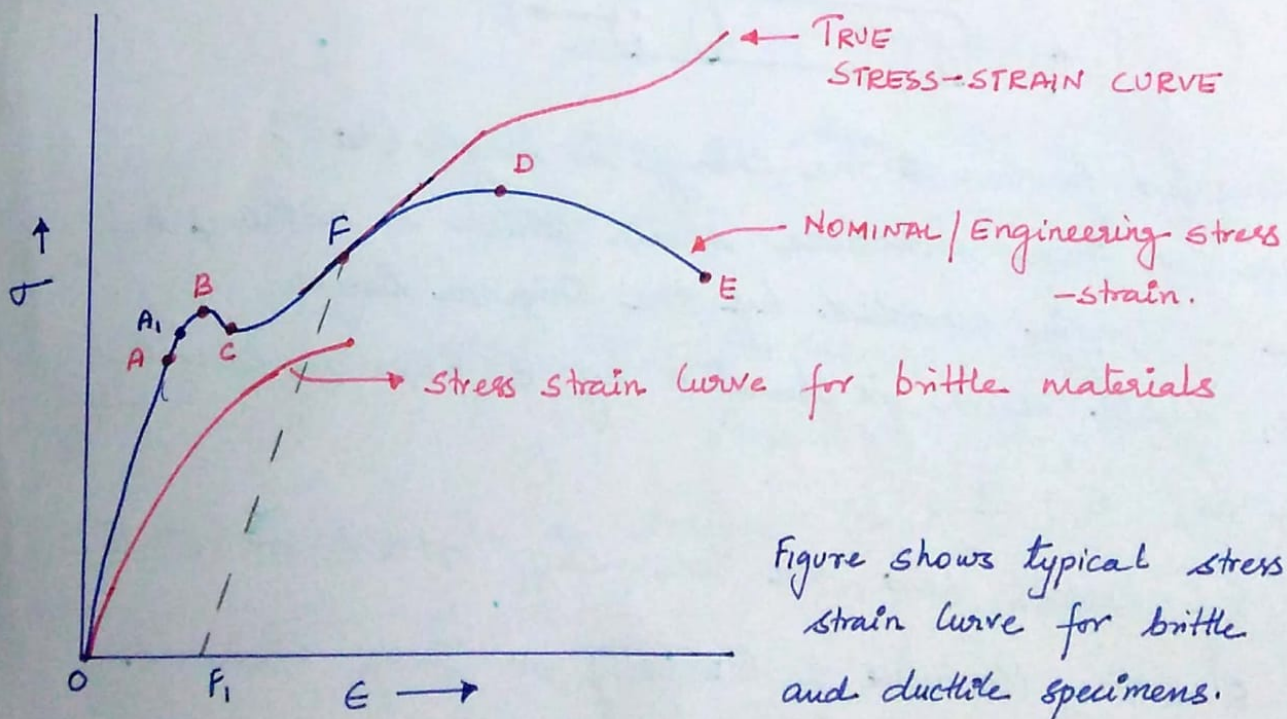


Figure shows typical stress strain curve for brittle and ductile specimens.

In the above figure, points A-E indicate the following. (3)

A → Proportionality Limit :- The point upto which stress remains proportional to strain.

A₁ → Elastic limit :- The point upto which, when it is unloaded, the specimen returns back to original shape without any permanent strain.

B → Upper yield point :- The point at which ~~stress~~^{load} decreases and strain increases. or at constant load there is a increase in strain.

C → Lower Yield point :- Here, the stress remains same but strain increases.

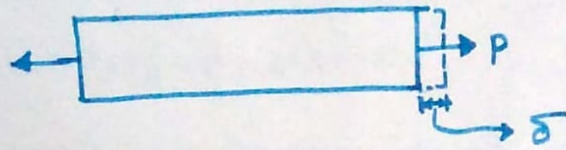
D → Ultimate stress :- This is the maximum stress the material can resist. At this point the cross section of the material decreases at a rapid rate.

E → Fracture point :- The stress at which the specimen finally fails.

F-F' → Permanent deformation :- In case the specimen is loaded beyond elastic limit and is unloaded the a straight-line parallel to the elastic limit is drawn and OF₁ is the permanent strain.

GENERALIZED HOOKE'S LAW:-

(4)

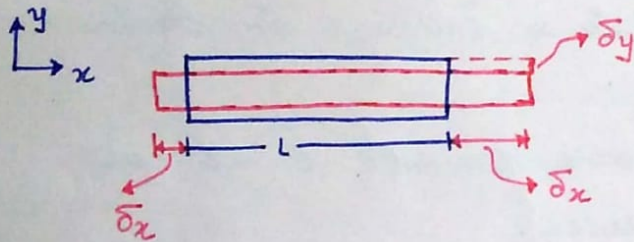


Consider the bar under Uniaxial tension, let the deflection be ' δ ' and ' σ ' be stress and ' ϵ ' be the strain with young's modulus E .

From Hooke's law, we can say,

$$\sigma = E \epsilon$$

Now, let us consider the 2-D case.



Now, we can see from the diagram, the elongation in the x -direction has induced a shortening in the y -direction.

We name ϵ_y as the lateral strain as it is $\perp^{\text{th}}$ to the direction of force and ϵ_x as the longitudinal strain.

The ratio of lateral strain by longitudinal gives us Poisson's ratio.

$\mu = -\frac{\epsilon_y}{\epsilon_x}$, we denote ϵ_y as -ve as it is reduction in dimension.

Let us consider that ϵ_y is due to σ_y (key assumptions)

$$\sigma_x = E \epsilon_x \quad \text{Similarly} \quad \sigma_y = E \epsilon_y$$

$$\text{We can re-write, as, } \epsilon_x = \frac{\sigma_x}{E}, \text{ and } \epsilon_y = \frac{\sigma_y}{E}$$

$$\text{Hence, we can say, } \epsilon_x = \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} \text{ for a 2-D case.}$$

Similarly for a 3-D situation.

$$\epsilon_x = \frac{\sigma_x}{E} - \frac{\mu}{E} \sigma_y - \frac{\mu}{E} \sigma_z \quad \text{--- (1)}$$

Similarly for other directions.

$$\epsilon_y = \frac{\sigma_y}{E} - \frac{\mu}{E} \sigma_x - \frac{\mu}{E} \sigma_z \quad \text{--- (2)}$$

$$\epsilon_z = \frac{\sigma_z}{E} - \frac{\mu}{E} \sigma_x - \frac{\mu}{E} \sigma_y \quad \text{--- (3)}$$

Writing the above three equations in Matrix form.

$$\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \end{bmatrix} = \begin{bmatrix} 1/E & -\mu/E & -\mu/E \\ -\mu/E & 1/E & -\mu/E \\ -\mu/E & -\mu/E & 1/E \end{bmatrix} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{bmatrix}$$

Hence, we can also write shear strains as

$$\left. \begin{aligned} \tau_{xy} &= G \gamma_{xy} \\ \tau_{yz} &= G \gamma_{yz} \\ \tau_{xz} &= G \gamma_{xz} \end{aligned} \right\} \text{for an isotropic material}$$

$\tau_{xy} = G \gamma_{xy} \rightarrow G \text{ - remains same in all directions.}$

$$\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{xz} \end{bmatrix} = \underbrace{\begin{bmatrix} 1/E & -\mu/E & -\mu/E & 0 & 0 & 0 \\ -\mu/E & 1/E & -\mu/E & 0 & 0 & 0 \\ -\mu/E & -\mu/E & 1/E & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G \end{bmatrix}}_{\text{Compliance Tensor}} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{xz} \end{bmatrix}$$

$\hat{\epsilon} \qquad \qquad \qquad \hat{\sigma}$

(6)

$\hat{\epsilon} = S \hat{\sigma}$, where S is the Compliance tensor or the Compliance matrix.

On Inversion of the matrix, we get.

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{xz} \end{bmatrix} = \begin{bmatrix} \frac{E}{1-\mu} & \mu & \mu & 0 & 0 & 0 \\ \mu & \frac{E}{1-\mu} & \mu & 0 & 0 & 0 \\ \mu & \mu & \frac{E}{1-\mu} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{E(1-2\mu)}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{E(1-2\mu)}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{E(1-2\mu)}{2} \end{bmatrix} \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{xz} \end{bmatrix}$$

$\hat{\sigma} = k \hat{\epsilon}$ where k is the stiffness matrix

and $\underline{E} = \frac{E}{(1+\mu)(1-2\mu)}$.

Extending it to a Anisotropic Material.

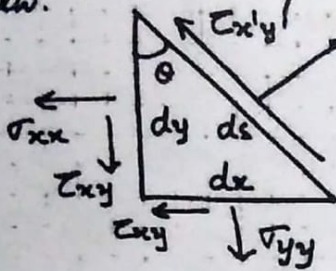
$$\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{xz} \end{bmatrix} = \begin{bmatrix} 1/E_x & -\mu_{xy}/E_y & -\mu_{xz}/E_z & 0 & 0 & 0 \\ -\mu_{yx}/E_x & 1/E_y & -\mu_{yz}/E_z & 0 & 0 & 0 \\ -\mu_{xy}/E_x & -\mu_{yz}/E_y & 1/E_z & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G_{xy} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G_{yz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G_{xz} \end{bmatrix} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{xz} \end{bmatrix}$$

Principal stresses and Principal strains:- (Analysis of stress and strain at a point).

The state of stress at a point in stressed body remains the same. But, the shear stress component vary as the orientation of plane through that point changes.

Under combined loading / complex loadings the member may experience larger stresses on inclined plane than on the cross sections.

It is important to know the max stress from failure point of view.



Considering thickness of the lamina to be 't'.

$\sigma_{x'y'}$ and τ_{xy} are normal & shear stresses acting along the plane, at angle θ .

Under equilibrium conditions, we get.

$$\sum F_{x'} = 0 \quad \sigma_{x'y'} \cdot t \cdot ds = -\sigma_{xx} \cdot t \cdot dy \cos\theta - \sigma_{yy} \cdot t \cdot dx \sin\theta - \tau_{xy} \cdot t \cdot dy \sin\theta - \tau_{xy} \cdot t \cdot dx \cos\theta = 0$$

dividing the equation by $t \cdot ds$ and considering $\frac{dy}{ds} = \cos\theta$ and $\frac{dx}{ds} = \sin\theta$ from the fig., we get,

$$\sigma_{x'y'} = \sigma_{xx} \cos^2\theta + \sigma_{yy} \sin^2\theta + 2\tau_{xy} \sin\theta \cos\theta \quad \text{--- (1)}$$

Similarly from the equilibrium condition, $\sum F_{y'} = 0$, we get,

$$\tau_{x'y'} = (\sigma_{yy} - \sigma_{xx}) \sin\theta \cos\theta + \tau_{xy} (\cos^2\theta - \sin^2\theta) \quad \text{--- (2)}$$

Using the trigonometry identities; $\cos^2\theta = \frac{1+\cos 2\theta}{2}$, $\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$\sin\theta \cos\theta = \frac{\sin 2\theta}{2}$, Substituting back into eqn (1) and (2), we get,

$$\sigma_{x'y'} = \sigma_{xx} \left(\frac{1+\cos 2\theta}{2} \right) + \sigma_{yy} \left(\frac{1-\cos 2\theta}{2} \right) + 2\tau_{xy} \frac{\sin 2\theta}{2}$$

$$\sigma_{x'y'} = \frac{\sigma_{xx} + \sigma_{yy}}{2} + \frac{(\sigma_{xx} - \sigma_{yy}) \cos 2\theta}{2} + \tau_{xy} \sin 2\theta \quad \text{--- (3)}$$

Similarly, we get

$$\tau_{x'y'} = \frac{(\sigma_{yy} - \sigma_{xx}) \sin 2\theta}{2} + \tau_{xy} \cos 2\theta \quad \text{--- (4)}$$

for stresses on the $y'y'$ face, substitute $\theta + 90^\circ$ for θ ;

$$\sigma_{y'y'} = \frac{\sigma_{xx} + \sigma_{yy}}{2} - \frac{(\sigma_{xx} - \sigma_{yy}) \cos 2\theta}{2} - \tau_{xy} \sin 2\theta \quad \text{--- (5)}$$

Summing the expressions (3) and (5), we get, $\sigma_{x'y'} + \sigma_{y'y'} = \sigma_{xx} + \sigma_{yy}$.

8

By taking the derivative of $\tau_{x'y'}$ with respect to θ , we get,

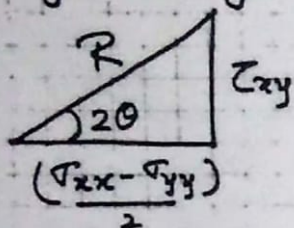
$$\frac{d\tau_{x'y'}}{d\theta} = -(\tau_{xx} - \tau_{yy}) \sin 2\theta + 2\tau_{xy} \cos 2\theta = 0$$

Equating the equation to zero,

$$0 = (\tau_{yy} - \tau_{xx}) \sin 2\theta + 2\tau_{xy} \cos 2\theta \quad \text{--- (6) Rearranging Eqn. (6)}$$

we get, $\tan 2\theta = \frac{2\tau_{xy}}{\tau_{xx} - \tau_{yy}} \quad \text{--- (7)}$

By drawing a triangle of Eqn. (7), we get,



From the triangle, we get,

$$R^2 = \left[\frac{\tau_{xx} - \tau_{yy}}{2} \right]^2 + \tau_{xy}^2 \quad \text{--- (a)}$$

$$\cos 2\theta = \frac{\tau_{xx} - \tau_{yy}}{2R} \quad \text{--- (b) and } \sin 2\theta = \frac{\tau_{xy}}{R} \quad \text{--- (c)}$$

Shear is max at 45°

Substituting Eqn. (a), (b) and (c) in Eqn. (5), we get,

$$\sigma_1 = \frac{\tau_{xx} + \tau_{yy}}{2} + \frac{\tau_{xx} - \tau_{yy}}{2} \left[\frac{\tau_{xx} - \tau_{yy}}{2R} \right] + \tau_{xy} \left[\frac{\tau_{xy}}{R} \right]$$

$$\rightarrow \sigma_1 = \frac{\tau_{xx} + \tau_{yy}}{2} + \sqrt{\left(\frac{\tau_{xx} - \tau_{yy}}{2} \right)^2 + \tau_{xy}^2} \quad \text{Similarly, we can write for } \sigma_2.$$

$$\rightarrow \sigma_2 = \frac{\tau_{xx} + \tau_{yy}}{2} - \sqrt{\left(\frac{\tau_{xx} - \tau_{yy}}{2} \right)^2 + \tau_{xy}^2}$$

Note: The shear stresses are zero on the principal planes.

To get max value of shear stress, the derivative of $\tau_{x'y'}$ w.r.t to θ is equated to zero.

$$\frac{d\tau_{x'y'}}{d\theta} = (\tau_{yy} - \tau_{xx}) \cos 2\theta - 2\tau_{xy} \sin 2\theta = 0.$$

Rearranging we get, $\tan 2\theta(s) = \frac{(\tau_{yy} - \tau_{xx})}{2\tau_{xy}} \quad \left[\begin{array}{l} \text{Subscript 's' denotes} \\ \text{Shear plane} \end{array} \right] \quad \text{--- (8)}$

Comparing eqn. (7) and (8).

$$\tan 2\theta = -\frac{1}{\tan 2\theta(s)}$$

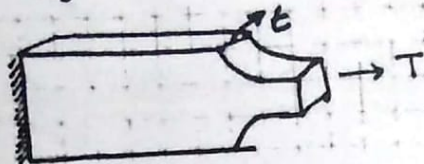
Substituting θ_s value in eqn. (5), we get

$$\tau_{\max} = \sqrt{\left(\frac{\tau_{xx} - \tau_{yy}}{2} \right)^2 + \tau_{xy}^2} = \frac{\sigma_{\max} - \sigma_{\min}}{2} \quad \parallel$$

Plane stress and Plane strain Approximations:-

Many problems in elasticity may be treated by a plane theory of elasticity.

Plane stress:- It is defined as a state of stress, in which the normal stress σ_z , and the corresponding shear stress τ_{xz} and τ_{yz} are assumed to be zero.



loading and Boundary Conditions
 (a) loadings may be point forces distributed forces applied over the thickness of the plate as shown.

(b) Supports may be fixed points/Edges or roller supports.

Hence for an isotropic material,

$$\tau_{xz} = \tau_{yz} = \tau_{yz} = 0, \text{ so, } \gamma_{xz} = \gamma_{yz} = 0$$

Writing the stiffness form of the generalized hooke's law we get

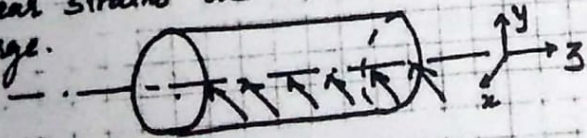
$$[\sigma] = [D][\epsilon] \text{ hence } D = \frac{\text{Stress matrix}}{\text{Strain}}$$

$$D = \begin{bmatrix} 1 & \mu & 0 \\ \mu & 1 & 0 \\ 0 & 0 & \frac{1-\mu}{2} \end{bmatrix} \frac{E}{1-\mu^2}$$

hence the compliance matrix reduces to

$$\begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} \frac{1}{E} & -\frac{\mu}{E} & 0 \\ -\frac{\mu}{E} & \frac{1}{E} & 0 \\ 0 & 0 & \frac{1}{G} \end{bmatrix} \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix}$$

Plane strain:- Defined as a state of strain in which the strain normal to xy plane and the corresponding shear strains are assumed to be zero when the thickness is very large.



Hence for an isotropic material,
 $\epsilon_z = \gamma_{xz} = \gamma_{yz} = 0$, hence
 $\tau_{xz} = \tau_{yz} = 0$

Hence, $[\sigma] = D[\epsilon]$

$$D = \begin{bmatrix} 1-\mu & \mu & 0 \\ \mu & 1-\mu & 0 \\ 0 & 0 & \frac{1-2\mu}{2} \end{bmatrix} \frac{E}{(1+\mu)(1-2\mu)}$$

State of stress in 3-D.

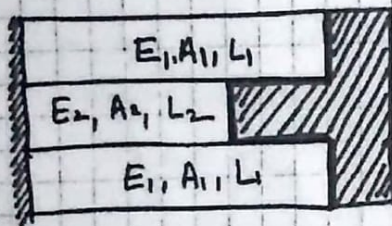
Case 1: If $\sigma_1 \neq \sigma_2 \neq \sigma_3 \neq 0 \rightarrow$ triaxial state of stress

Case 2: If $\sigma_1 \neq \sigma_2 = \sigma_3 \rightarrow$ cylindrical state of stress.

Case 3: If $\sigma_1 = \sigma_2 = \sigma_3 \rightarrow$ hydrostatic state of stress.

Mechanics of Compound bar Under axial loading:-

Now, let us consider a compound bar with two different material properties, different cross-sectional areas and also two different lengths as shown in the figure.



Let us consider the condition for static equilibrium
 $P = P_1 + P_2 + P_1 \quad \text{--- (1)}$

The load acting over the entire body is summation of load acting on the individual elements.

Now, let us consider the compatibility condition,

(2) $\leftarrow \Delta_1 = \Delta_2$, the deflection in the first body equals the deflection in the second. From (1) and (2) we get,

$$\frac{P_1 L_1}{A_1 E_1} = \frac{P_2 L_2}{A_2 E_2} \quad \text{--- (3)} \quad \text{So from (1) and (3) we}$$

Can solve for P_1 and P_2 , And hence find the stresses induced in each element. $\frac{P_1}{A_1} = \sigma_1$ and $\frac{P_2}{A_2} = \sigma_2$. And the Overall deflection in the compound bar due to the load P .

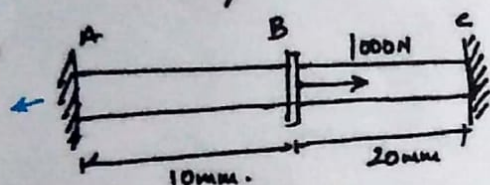
Hyper static Systems or otherwise termed statically indeterminate structures/systems

(11)

In this type of system, the number of Unknown reactions are greater than the number of equations available to solve the system.

Also, in addition to the equilibrium equations, we need one or more compatibility equations to solve for the reactions.

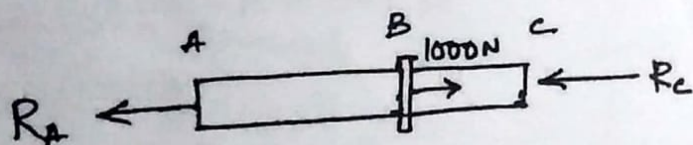
A Simple Example:-



Both Ends are fixed, with the axial force ~~and~~ applied at B.

Now, the three equilibrium equations are $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$

We see that the free body diagram below shows there



will not be any moments as the forces all pass through the same line and there are no forces

in the y-direction and hence we are left with.

$$\sum F_x = 0 = -R_A - R_C + 1000 = 0, \text{ hence}$$

$R_A + R_C = 1000$, we have two Unknown's but one equation.

But, because both ends are fixed, we can say, that the overall length change is zero. So $\Delta_{AB} + \Delta_{BC} = 0$.

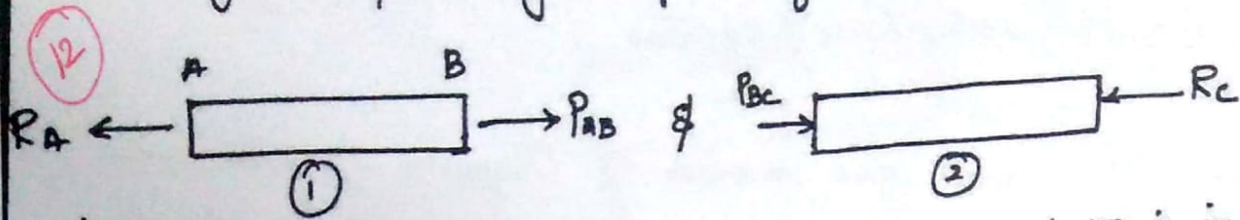
$$\therefore \frac{P_{AB} L_{AB}}{EA} + \frac{P_{BC} L_{BC}}{EA} = 0$$

$$\Rightarrow P_{AB}(10) + P_{BC}(20) = 0$$

$$P_{AB} + 2P_{BC} = 0 \quad \text{--- (1)}$$

Now, we have to relate the force to the reactions.

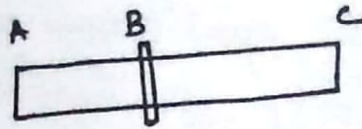
Drawing the free body diagram of the individual segments.



Hence from ① we can say. (Note, Segment AB is in tension and BC is in compression).

$$R_A = P_{AB}$$

and $P_{BC} = -P_{AB}$



~~or $P_{BC} = -P_{AB}$~~

Hence from Equ ①, we get.

$$P_{AB} + 2P_{BC} = 0, \text{ Now, we know, } R_A = P_{AB}$$

Replacing $P_{BC} = R_A - 1000 = P_{BC}$
 $R_A = P_{AB}$

$$R_A + 2(R_A - 1000) = 0$$

$$R_A + 2R_A - 2000 = 0$$

$$3R_A = 2000$$

$$R_A = 666.667 \text{ N.}$$

And hence $R_A + P_C = 1000$
 $P_C = 1000 - 666.67$
 $P_C = 333.33 \text{ N.}$

$$R_A + R_C = 1000$$

$$R_C = 1000 - R_A$$

Now, we see that R_C is compressive, hence R_C becomes -ve.

$$\therefore P_C = (R_A - 1000)$$

(Note: Principle of Superposition continues).

Principle of Superposition:-

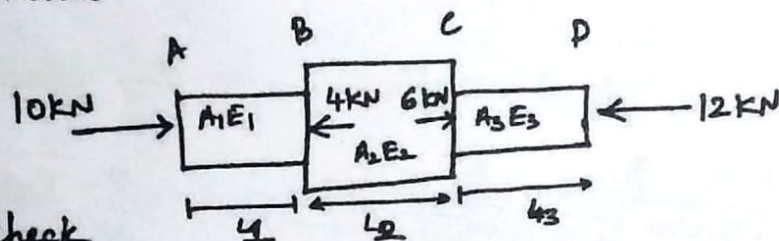
(13)

for a system of bars with varying cross-section we can apply the principle of Superposition.

Which is,

$$\Delta = \sum \Delta = \sum_{i=1}^N \frac{PL}{AE}$$

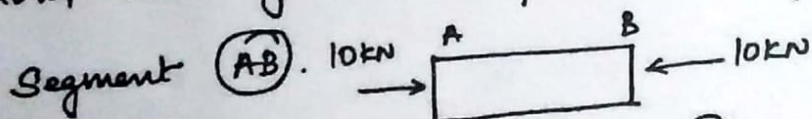
Now for a given bar/system below, we can find the total deformation by adding the individual deformation of the bar segments.



Now, let us check if the system satisfies the static equilibrium conditions.

$$\sum F_x = 0, \quad +10 - 4 + 6 - 12 = 0.$$

Now, drawing the FBD for each segment.

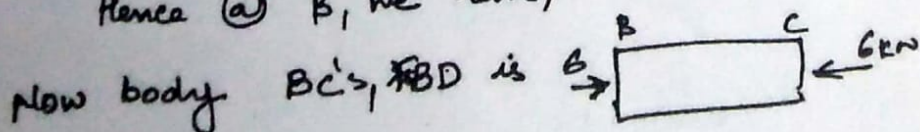


Due to the compressive load at (A) we have the compressive reaction at B.

Now at junction (B) we have,

-10kN (compressive) and +4kN which is tension for the Segment-BC.

Hence at B, we have, $-10 + 4 = -6$ hence compressive.

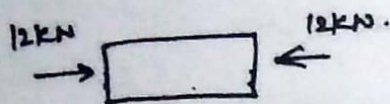


14

Now, at junction (C) we have,

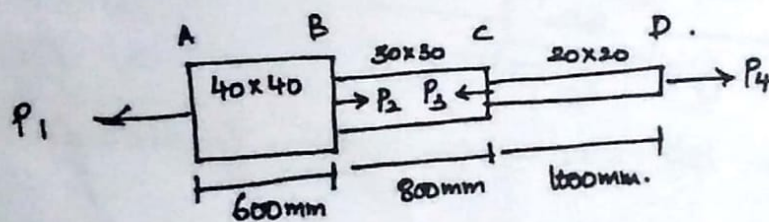
- 6 kN compressive and 6 kN ^{Comp} ~~tensile~~ acting on CD

hence, $-6 - 6 = -12$ Compressive and hence the segment CD has a Compressive force 12 kN acting on it.



Hence $\Delta_s = \sum PL/AE$. Where P varies from each segment.

Example:- Find the total deflection for the following system:-



$$P_1 = 80 \text{ kN}$$

$$P_2 = 60 \text{ kN}$$

$$P_3 = 40 \text{ kN}$$

$$P_4 = ?$$

$$\text{taking } E = 2 \times 10^5 \text{ N/mm}^2$$

$$P_4 = 60 \text{ kN}$$

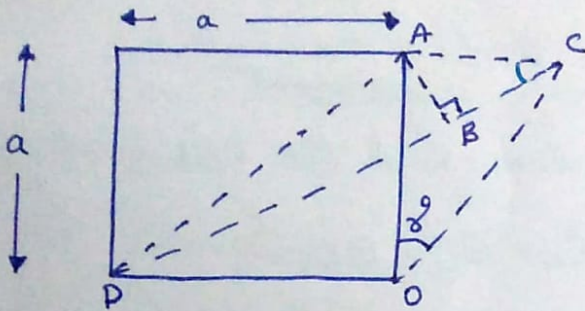
$$\Delta_1 = 0.15 \text{ mm}$$

$$\Delta_2 = 0.0889 \text{ mm}$$

$$\Delta_3 = 0.75$$

$$\Delta_T = \Delta_1 + \Delta_2 + \Delta_3 = \underline{0.9889 \text{ mm}}$$

Relationship b/w Elastic Constants:-



$$\angle ABC = 90^\circ$$

$$\angle AOC = \gamma \text{ (Small shear strain)}$$

$$\angle ADO = \angle CDO = \angle BCA \text{ (Small strain)} \\ = 45^\circ$$

As AB is $\perp$ to DC, we can say $AD = BD$, hence, the strain on the diagonal is,

$$\frac{BC}{DB} = \frac{BC}{DA} = \frac{AC \cos 45^\circ}{a\sqrt{2}} = \frac{AC \sqrt{2}}{a\sqrt{2} \cdot 2} = \frac{AC}{2a} \quad \text{--- (1)}$$

From the above figure, we can say, since γ is very small.

$$\frac{AC}{AO} = \gamma, \quad AC = \gamma \cdot AO \Rightarrow AC = \gamma \cdot a.$$

Substituting back in (1).

$$\frac{\gamma \cdot a}{2a} = \text{strain in diagonal AD} = \gamma/2$$

$$\epsilon_{(AD)} = \gamma/2, \quad \text{From Hooke's Law, } \tau = G\gamma \quad \text{or } \frac{\gamma}{2} = \frac{\tau}{G} = 2\epsilon_{AD}$$

For a 2-D case,

$$\epsilon_{(AD)} = \frac{\tau}{E} - \frac{\mu(-\tau)}{E} = \frac{\tau}{E} + \frac{\mu\tau}{E} \quad \text{--- (1)}$$

$$\text{Hence, } \epsilon_{AD} = \frac{\tau}{2G} \quad \text{--- (2)}, \quad \text{From (1) \& (2), we get,}$$

$$\frac{\tau}{2G} = \frac{\tau}{E} + \frac{\mu\tau}{E}, \quad \frac{\tau}{2G} = \frac{\tau}{E} [1 + \mu], \quad E = 2G [1 + \mu].$$

Relationship b/w E , K , & μ :-

Volumetric strain, $K = \frac{P}{dv}$, where $dv = \epsilon_x + \epsilon_y + \epsilon_z$
and P is the applied body force, and ' K ' is the bulk modulus,
Considering no change in shape, $\epsilon_x = \epsilon_y = \epsilon_z$,

Similarly hook's law becomes,

$$\epsilon_{xx} = \frac{P}{E} - \frac{\mu P}{E} - \frac{\mu P}{E} = \frac{P}{E} [1 - 2\mu]$$

Similarly, $\epsilon_{yy} = P/E [1 - 2\mu]$ and $\epsilon_{zz} = P/E [1 - 2\mu]$

hence, Volumetric becomes,

$$dv = \frac{3P}{E} [1 - 2\mu]$$

hence bulk modulus becomes, $K = \frac{P}{\frac{3P}{E} [1 - 2\mu]}$

$$K = \frac{E}{3[1 - 2\mu]}$$

hence, we have,

$$E = 2G[1 + \mu] \quad \text{--- (3)}$$

$$E = 2K[1 - 2\mu] \quad \text{--- (4)}$$

hence solving (3) and (4), we get,

$$E = \frac{9GK}{(3K + G)}$$

$$\text{or } \mu = 0.5 [3K - 2G] [G + 3K]. //$$

Thermal stresses:-

When the temperature rises materials tend to expand and when the temperature drops materials contract.

The change in length can be denoted as,

$\Delta = \alpha t L$, where α is coefficient of thermal expansion (change in Unit length due to change in Unit temperature).

$$\Delta = \alpha t L \quad \text{or}$$

$$\frac{PL}{AE} = \alpha t L, \quad \text{Simplifying,}$$

$$\frac{P}{A} = E \alpha t \quad //$$

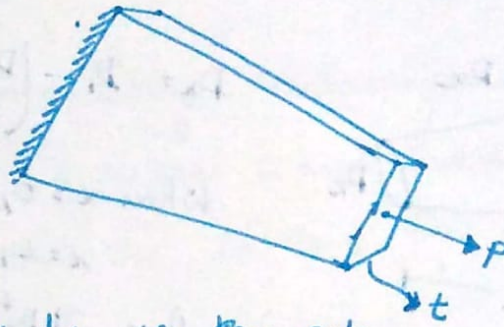
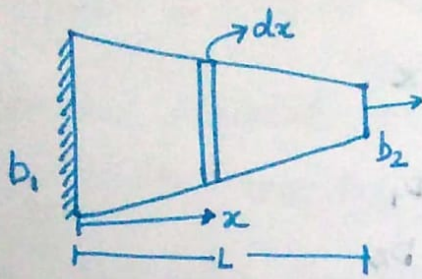
Stress Tensor:-

Due to the force F exerted on a solid body, the body experiences 9-components of stresses, Normal and tangential or direct and shear stresses.

Hence a stress tensor can be represented by,

$$\sigma_{ij} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} \quad //$$

Tapered bar of Uniform thickness:-



to find the width of element dx , we can get

$$b_x = b_1 - \left(\frac{b_1 - b_2}{L} \right) x, \text{ hence, when } x=0, b_x = b_1 \text{ and } x=L, b_x = b_2$$

We assume $k = \frac{b_1 - b_2}{L}$, and the cross sectional for element will be,

$$(b_1 - kx)xt = A_x, \text{ hence stress is given as;}$$

$$\sigma = \frac{P}{A_x} = \frac{P}{(b_1 - kx)t} \quad \text{and strain } \epsilon = \frac{dx}{dx} \text{ or } dx \cdot \epsilon = dl_x$$

$$\epsilon = \frac{\sigma}{E}, \therefore dx \cdot \frac{P}{(b_1 - kx)t \cdot E} = dl_x$$

Integrating we get, (along the entire length)

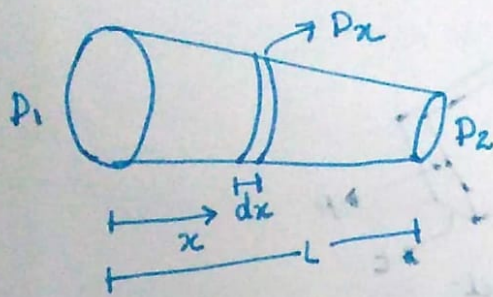
$$dl_x = \int_0^L \frac{P}{(b_1 - kx)t \cdot E} \cdot dx \Rightarrow \frac{P}{t \cdot E} \int_0^L \frac{1}{(b_1 - kx)} \cdot dx$$

$$dl = \frac{P}{t \cdot E} \left[\log(b_1 - kx) \right]_0^L \cdot \left[\frac{-1}{-k} \right] = \frac{-P}{t \cdot E \cdot k} \left[\log(b_1 - kx) \right]_0^L$$

$$dl = \frac{-P}{t \cdot E \cdot k} \left[\log(b_1 - kL) - \log(b_1) \right] = \frac{P}{t \cdot E \cdot k} \left[\log(b_1) - \log(b_1 - kL) \right]$$

$$dl = \frac{P}{t \cdot E \left[\frac{b_1 - b_2}{L} \right]} \left[\log \frac{b_1}{b_1 - \frac{b_1 - b_2}{L} \cdot L} \right] = dl = \frac{PL}{t \cdot E [b_1 - b_2]} \left[\log \frac{b_1}{b_2} \right]$$

Uniformly Tapered Circular Bar:-



$$D_x = D_1 - \left(\frac{D_1 - D_2}{L} \right) \cdot x$$

$$\text{When } x = 0, D_x = D_1$$

$$x = L, D_x = D_2$$

$$A_x = \frac{\pi D_x^2}{4} = \frac{\pi}{4} [D_1 - kx]^2$$

$$\sigma = \frac{P}{A_x} = \frac{4P}{\pi [D_1 - kx]^2} \quad \therefore dx \cdot \epsilon = dl$$

$$dl = \frac{4P}{E\pi (D_1 - kx)^2} \cdot dx, \quad \text{Integrating Over the entire length,}$$

$$dl = \frac{4P}{E\pi} \int_0^L \frac{1}{(D_1 - kx)^2} \cdot dx, \quad \Rightarrow \frac{4P}{E\pi} \left[\frac{1}{D_1 - kx} \right]_0^L$$

$$dl = \frac{4P}{E\pi} \left[\frac{-1}{(D_1 - kx)} \cdot \left[\frac{1}{-k} \right] \right]_0^L \Rightarrow \frac{4P}{E\pi k} \left[\frac{1}{D_1 - kx} \right]_0^L$$

$$dl = \frac{4PL}{E\pi [D_1 - D_2]} \left[\frac{1}{D_1 - \left(\frac{D_1 - D_2}{L} \right) x} \right]_0^L = \frac{4PL}{E\pi (D_1 - D_2)} \left[\frac{1}{D_1 - \frac{(D_1 - D_2)L}{L}} - \frac{1}{D_1 - \frac{(D_1 - D_2)0}{L}} \right]$$

$$dl = \frac{4PL}{E\pi [D_1 - D_2]} \left[\frac{1}{D_2} - \frac{1}{D_1} \right]$$

$$dl = \frac{4PL}{E\pi [D_1 - D_2]} \left[\frac{D_1 - D_2}{D_1 D_2} \right]$$

$$dl = \frac{4PL}{E\pi D_1 D_2} //$$

Examples on Tapered bar's:-

- ① Bone Material has a stress strain relationship as follows, $\sigma = E \left[\epsilon / (1 + K E \epsilon) \right]$ where K and E are constants.

Determine compression within the length of the bone assuming the cross sectional area is constant.

Sol:- $\sigma = P/A$ & $\epsilon = \frac{dL_x}{dx}$



$$\sigma = E \epsilon$$

or

$$\sigma = E \left[\frac{\epsilon}{1 + K E \epsilon} \right]$$

$$\sigma = \frac{E \epsilon}{1 + K E \epsilon} \quad \text{Substituting for } \epsilon, \quad \sigma = \frac{E (dL_x/dx)}{1 + K E [dL_x/dx]} = \frac{P}{A}$$

$$\frac{P}{A} \left[1 + K E \left[\frac{dL_x}{dx} \right] \right] = E \left(\frac{dL_x}{dx} \right)$$

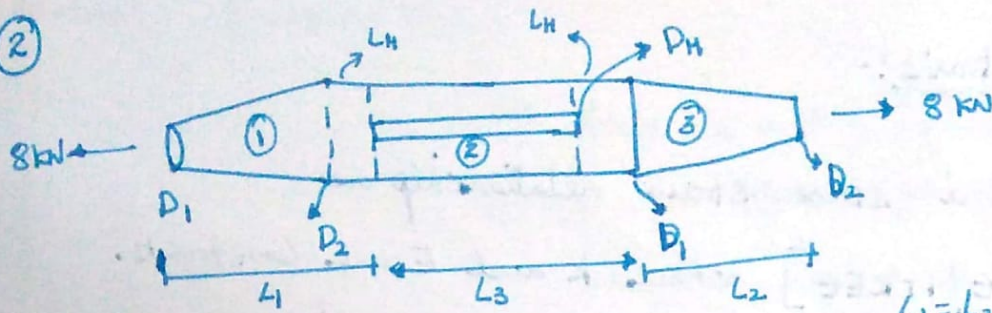
$$\frac{P}{A} + \frac{P K E}{A} \left[\frac{dL_x}{dx} \right] = E \left[\frac{dL_x}{dx} \right]$$

$$\frac{P}{A} = E \left[\frac{dL_x}{dx} \right] - \frac{P K E}{A} \left[\frac{dL_x}{dx} \right] \Rightarrow E \left[\frac{dL_x}{dx} \right] \left[1 - \frac{K P}{A} \right]$$

$$\frac{P}{A} = \left[\frac{dL_x}{dx} \right] \left[E - \frac{P K E}{A} \right] \Rightarrow \int_0^L \frac{P \cdot dx}{A \left[E - \frac{P K E}{A} \right]} = dL_x, \text{ Integrating,}$$

$$dL_x = \frac{P L}{A E \left[1 - \frac{P K}{A} \right]}, \quad dL_x = \frac{P L}{E [A - P K]} //$$

②



$$E = 70 \text{ GPa}$$

$$D_1 = 10 \text{ mm}$$

$$D_2 = 16 \text{ mm}$$

$$D_h = 2.5 \text{ mm}$$

$$L_1 = L_2 = 30 \text{ mm}$$

$$L_3 = 50 \text{ mm}$$

$$L_h = 20 \text{ mm}$$

Sol:- $\delta = \frac{4PL}{E\pi d_1 d_2}$ $\delta_2 = \frac{PL}{AE}$

$$\delta_{1,3} = \frac{4 \times 8 \times 10^3 \times 30}{0.7 \times 10^5 \times 3.14 \times 10 \times 16}$$

$$\delta_h = \frac{PL}{AE}, \quad A = \frac{\pi d_2^2}{4}$$

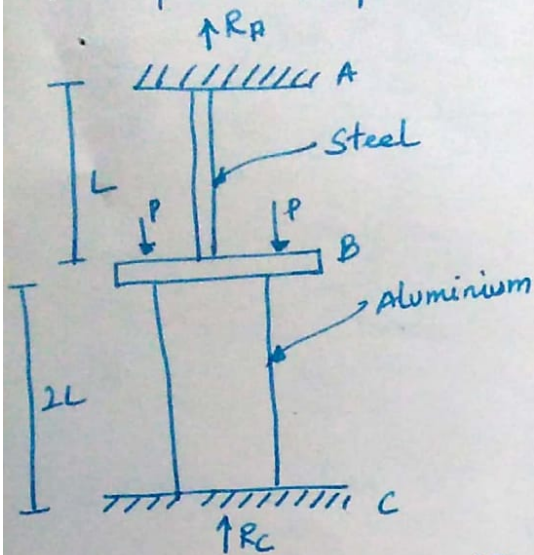
$$\delta_3 = \frac{PL}{AE}, \quad A = \frac{\pi}{4} (d_2^2 - d_h^2)$$

Total deformation will be

$$\delta_T = \delta_1 + \delta_h + \delta_2 + \delta_h + \delta_3$$

Substituting, we can solve for axial deformation.

Examples:- Compound Bars:-



- (a) Obtain σ_A and σ_S in steel & Aluminium.
 - (b) Calculate stresses for 12 kN force and Cross section of 10 mm^2 in Aluminium and 2.5 mm^2 in steel.
- Modulus of Elasticity is 70 GPa for Al and 210 GPa for steel.

Sol: $\sum F_y = 0, R_A + R_C = 2P$ — (1)

Compatibility Equations:-

$$\delta_{AC} = \delta_{BC} \text{ or } \delta_{AC} - \delta_{BC} = 0 \quad \text{--- (2)}$$

$$\delta_{AC} = \frac{R_A \cdot L}{A_S E_S} \quad \text{and} \quad \delta_{BC} = \frac{R_C \cdot 2L}{A_A E_A}$$

Substituting in (2)

$$\frac{R_A \cdot L}{A_S E_S} - \frac{R_C \cdot 2L}{A_A E_A} = 0 \quad \text{--- (3)}$$

From (1), we get $R_A = 2P - R_C$, substituting in (3),

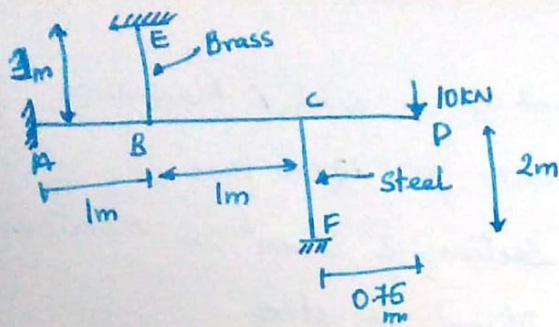
$$\frac{(2P - R_C) \cdot L}{A_S E_S} - \frac{R_C \cdot 2L}{A_A E_A} = 0$$

$$\text{or } L \frac{(2P - R_C)}{A_S E_S} = \frac{R_C \cdot 2L}{A_A E_A}$$

$$\frac{L(2P - R_C)}{2L \cdot R_C} = \frac{A_S E_S}{A_A E_A} \Rightarrow \frac{4(A_S E_S)}{A_A E_A} = \frac{2P}{R_C} - \frac{1}{0} \Rightarrow \frac{4(A_S E_S) + 1}{A_A E_A} = P/R_C$$

Rearranging, we get,

$$R_C = \frac{P (A_A E_A)}{4(A_S E_S) + A_A E_A}$$



$$\sum M_A = 0 \quad P_b \times 1 + P_s \times 2 = 10 \times 10^3 \times 2.75$$

$$P_b + 2P_s = 27500 \quad \text{--- (1)}$$

deflection of brass & steel. $\frac{\Delta_b}{1} = \frac{\Delta_s}{2}$

$$\Delta_s = 2 \Delta_b$$

$$\frac{P_s L_s}{A_s E_s} = 2 \left[\frac{P_b L_b}{A_b E_b} \right]$$

$$P_s = 1.2 P_b \quad \text{--- (2)}$$

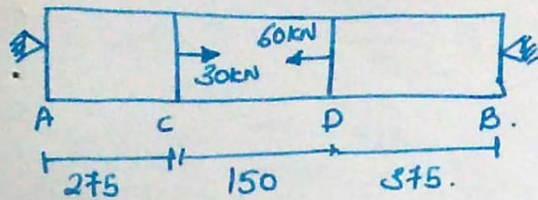
Solving (1) & (2),

$$P_b = 8088.24 \text{ N}$$

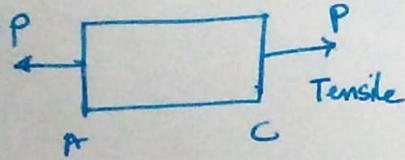
$$P_s = 9705.88 \text{ N}$$

$$\sigma_b = \frac{P_b}{A_b} = 8.08 \text{ N/mm}^2$$

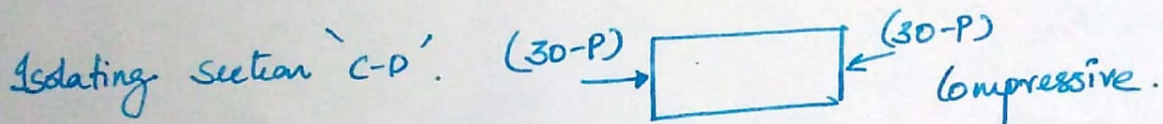
$$\sigma_s = \frac{P_s}{A_s} = 16.176 \text{ N/mm}^2$$



Isolating 'AC' Diameter of bar is 25mm.
Find stresses and change in each portion.



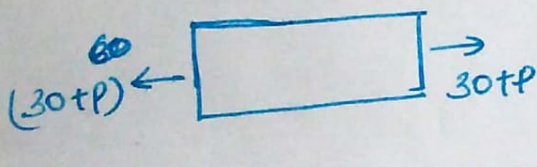
(a) Section C the load is tensile.



(a) Section C $(30-P)$.

and (a) Section D-B'.

but (a) Section D
60 Tensile - $(30-P)$ (comp).



$$60 - 30 + P \\ (30 + P)$$

But total deflection is zero, hence,

$$PL/AE]_1 + PL/AE]_2 + PL/AE]_3 = 0.$$

Solving we get P.